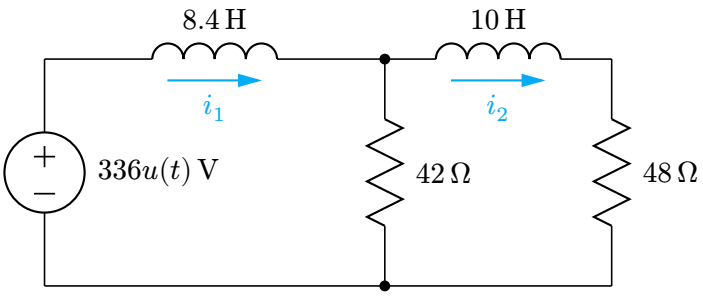
Lab 9: The Laplace Transform

ECE2200L Section 04, Fall 2026

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No 1.



Step 1. There is no initial energy stored in this circuit. Find $i_1(t)$ and $i_2(t)$ for $t > 0$. Use the LT approach.

$$i_1(0^+) = i_2(0^+) = 0\,\text{A}$$

$$8.4\frac{\mathrm{d}i_1}{\mathrm{d}t} + 42(i_1 - i_2) = 336$$

$$\overset{\mathcal{L}}{\Longrightarrow} \quad 8.4(sI_1 - \cancelto{0}{i_1(0^+)}) + 42(I_1 - I_2) = \frac{336}{s}$$

$$(8.4s + 42)I_1 - 42I_2 = \frac{336}{s}$$

$$10\frac{\mathrm{d}i_2}{\mathrm{d}t} + 48i_2 - 42(i_1 - i_2) = 0$$

$$\overset{\mathcal{L}}{\Longrightarrow} \quad 10(sI_2 - \cancelto{0}{i_2(0^+)}) + 48I_2 - 42(I_1 - I_2) = 0$$

$$-42I_1 + (10s + 90)I_2 = 0$$

from the second equation:

$$I_2 = \frac{42}{10s + 90}I_1$$

substitute into the first equation:

$$(8.4s + 42)I_1 - 42\left(\frac{42}{10s + 90}I_1\right) = \frac{336}{s}$$

$$[(8.4s + 42)(10s + 90) - 42^2]I_1 = \frac{336(10s + 90)}{s}$$

$$
\begin{aligned}
I_1 &= \frac{336(10s+90)}{s[(8.4s+42)(10s+90)-42^2]} \\
&= \frac{336(10s+90)}{s(84s^2+1176s+2016)} \\
&= \frac{336(10s+90)}{84s(s+2)(s+12)} \\
&= \frac{4(10s+90)}{s(s+2)(s+12)} \\
I_1 &= \frac{40(s+9)}{s(s+2)(s+12)}
\end{aligned}
$$

$$
I_2 = \frac{42}{\cancel{10s+90}}\left(\frac{\cancel{40(s+9)}^{\,4}}{s(s+2)(s+12)}\right)
$$

$$
I_2 = \frac{168}{s(s+2)(s+12)}
$$

$$
I_1 = \frac{A}{s} + \frac{B}{s+2} + \frac{C}{s+12}
$$

$$
40(s+9) = A(s+2)(s+12) + Bs(s+12) + Cs(s+2)
$$

$s=-2 \Rightarrow 280 = -20B \Rightarrow B = -14$

$s=-12 \Rightarrow -120 = 120C \Rightarrow C = -1$

$\Rightarrow A = 15$

$$
I_2 = \frac{A}{s} + \frac{B}{s+2} + \frac{C}{s+12}
$$

$$
168 = A(s+2)(s+12) + Bs(s+12) + Cs(s+2)
$$

$s=-2 \Rightarrow 168 = -20B \Rightarrow B = -8.4$

$s=-12 \Rightarrow 168 = 120C \Rightarrow C = 1.4$

$\Rightarrow A = 7$

$$
I_1 = \frac{15}{s} - \frac{14}{s+2} - \frac{1}{s+12}, \quad I_2 = \frac{7}{s} - \frac{8.4}{s+2} + \frac{1.4}{s+12}
$$

taking the inverse laplace transform for both:

$$
\boxed{
\begin{aligned}
i_1(t) &= 15 - 14e^{-2t} - e^{-12t}\,\mathrm{A} \quad , \quad \text{for } t \geq 0 \\
i_2(t) &= 7 - 8.4e^{-2t} + 1.4e^{-12t}\,\mathrm{A}\,, \quad \text{for } t \geq 0
\end{aligned}
}
$$

Step 2. Identify the forced and natural responses.

the forced responses are the steady-state constants:

$$i_{1f}(t) = 15\,\text{A}, \quad i_{2f}(t) = 7\,\text{A}$$

while the natural responses refer to the decaying exponential terms:

$$i_{1n}(t) = -14e^{-2t} - e^{-12t}, \quad i_{2n}(t) = -8.4e^{-2t} + 1.4e^{-12t}$$

Step 3. Simulate the circuit to confirm your results in Step 1.

View simulation code

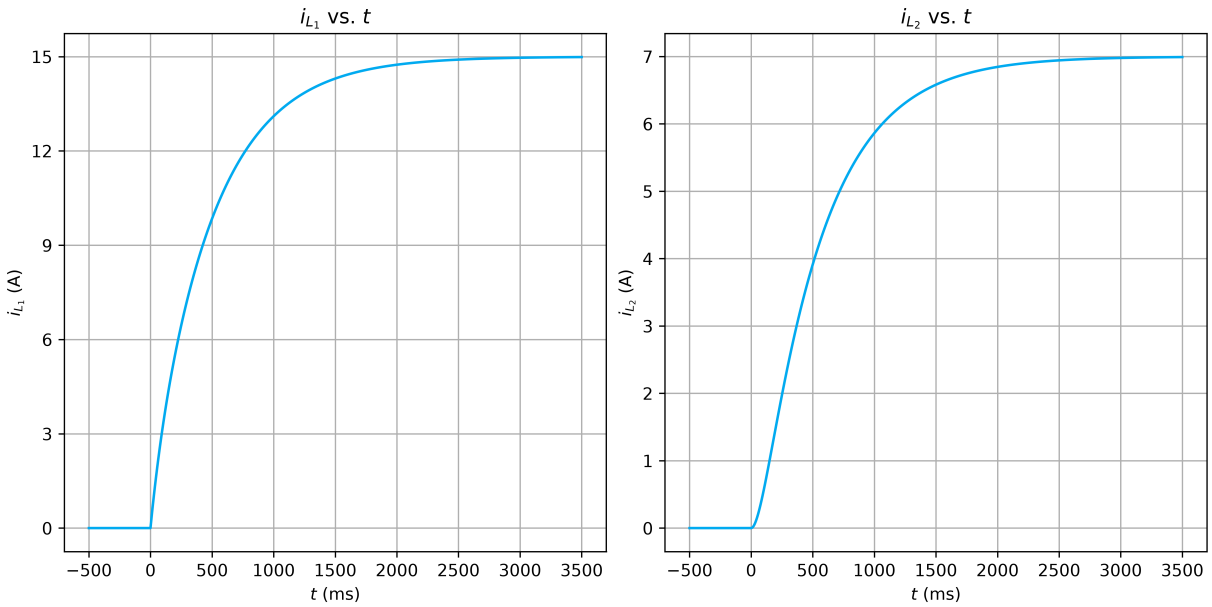


Figure 1: Simulated inductor currents $i_1(t)$ and $i_2(t)$ following application of the $336\,\text{V}$ step input. Both currents rise from zero and approach the manually calculated steady-state values of $15\,\text{A}$ and $7\,\text{A}$, respectively. The exponential rise closely matches the analytical solutions.

No 2.

$$F(s) = \frac{10s}{s^2 + 2s + 5}$$

Step 1. Find $f(t)$ for $t \geq 0$

$$\begin{aligned} F(s) &= \frac{10s}{s^2 + 2s + 1 + 5 - 1} \\ &= \frac{10s}{(s+1)^2 + 2^2} \\ &= \frac{10(s + 1 - 1)}{(s+1)^2 + 2^2} \\ &= 10\left[\frac{s+1}{(s+1)^2 + 2^2} - \frac{1}{(s+1)^2 + 2^2}\right] \\ &= 10\left[\frac{s+1}{(s+1)^2 + 2^2}\right] - 5\left[\frac{2}{(s+1)^2 + 2^2}\right] \end{aligned}$$

$$\overset{\mathcal{L}^{-1}}{\Longrightarrow} \quad \boxed{f(t) = 10e^{-t}\cos(2t) - 5e^{-t}\sin(2t)\,, \quad \text{for } t \geq 0}$$

No 3.

$$F(s) = \frac{70s - 268125}{(s + 875 - j3000)(s + 875 + j3000)}$$

Step 1.

$$F(s) = \frac{70s - 268125}{(s + 875)^2 + 3000^2}$$

rewrite numerator:

$$70s - 268125 = 70(s + 875) + x$$
$$x = -329375$$

$$F(s) = \frac{70(s + 875) - 329375}{(s + 875)^2 + 3000^2}$$

$$F(s) = 70\left[\frac{s + 875}{(s + 875)^2 + 3000^2}\right] - \frac{329375}{3000}\left[\frac{3000}{(s + 875)^2 + 3000^2}\right]$$

$$\overset{\mathcal{L}^{-1}}{\Longrightarrow} \quad f(t) = 70e^{-875t}\cos(3000t) - \frac{2635}{24}e^{-875t}\sin(3000t), \quad \text{for } t \geq 0$$

$$\boxed{f(t) = e^{-875t}\left[70\cos(3000t) - \frac{2635}{24}\sin(3000t)\right], \quad \text{for } t \geq 0}$$

No 4.

$$F(s) = \frac{768}{\left(s^2 + 6s + 25\right)^2} = \frac{768}{(s + 3 - j4)^2(s + 3 + j4)^2}$$

Step 1. Find the partial fraction expansion when there are repeated complex conjugate roots.

$$F(s) = \frac{A}{s + 3 - j4} + \frac{B}{(s + 3 - j4)^2} + \frac{C}{s + 3 + j4} + \frac{D}{(s + 3 + j4)^2}$$

let $p = -3 + j4$, $q = -3 - j4$

$$\frac{768}{(s - p)^2(s - q)^2} = \frac{A}{s - p} + \frac{B}{(s - p)^2} + \frac{C}{s - q} + \frac{D}{(s - q)^2}$$

$$\frac{768}{(s - q)^2} = A(s - p) + B + \frac{C(s - p)^2}{s - q} + \frac{D(s - p)^2}{(s - q)^2}$$

setting $s = p$:

$$\frac{768}{(p - q)^2} = B \Rightarrow B = \frac{768}{(j8)^2} \Rightarrow B = -12$$

differentiating both sides and setting $s = p$:

$$-\frac{1536}{(p-q)^3} = A \Rightarrow A = -\frac{1536}{(j8)^3} \Rightarrow A = -3j$$

C and D are obtained by taking the complex conjugates of A and B

$$C = A^* = 3j$$
$$D = B^* = -12$$

so,

$$\boxed{F(s) = -\frac{3j}{s+3-j4} - \frac{12}{(s+3-j4)^2} + \frac{3j}{s+3+j4} - \frac{12}{(s+3+j4)^2}}$$

No 5.

this is a duplicate of No 1, with the exception of an added Step 2, which is addressed in No 1.